# ML Challenge: Know Your Data

## KAUST–Mawhiba IOAI Training

ML Competition

KAUST Academy
King Abdullah University of Science and Technology

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# The Challenge

You will solve *two tasks* with *one combined score*:

### Task 1 — Classification

**Predict Customer Churn**

Will a telecom customer leave?

Target: $\texttt{Churn} \in \{0, 1\}$

### Task 2 — Regression

**Predict Insurance Charges**

How much will medical insurance cost?

Target: $\texttt{charges} \in \mathbb{R}^+$

$$\text{Score} = \frac{\text{Macro F1} + \max(0,\ R^2)}{2} \times 100$$

# Dataset 1: Telco Customer Churn

**Scenario:** A telecom company wants to predict which customers are likely to *cancel their service* (churn).

- Each row is one customer
- Features describe demographics, services, and billing
- Target: `Churn` — **0** = stayed, **1** = left

### Why does this matter?

Acquiring a new customer costs **5–25**$\times$ more than retaining an existing one. Predicting churn lets the company intervene early.

# Churn Dataset — Features (19 columns)

**Demographics**

- `gender`
- `SeniorCitizen`
- `Partner`
- `Dependents`

**Services**

- `PhoneService`
- `MultipleLines`
- `InternetService`
- `OnlineSecurity`
- `OnlineBackup`
- `DeviceProtection`
- `TechSupport`
- `StreamingTV`
- `StreamingMovies`

**Billing**

- `tenure`
- `Contract`
- `PaperlessBilling`
- `PaymentMethod`
- `MonthlyCharges`
- `TotalCharges`

15 categorical + 2 integer + 2 float features

# Churn Dataset — Key Statistics

| **Property** | **Value** |
|---|---|
| Training samples | 5,625 |
| Test samples | 1,407 |
| Number of features | 19 |
| Churn rate (train) | *26.6%* |

Watch out — Class Imbalance!

Only ~1 in 4 customers churned. A model that always predicts "no churn" gets 73% accuracy but **terrible Macro F1**.

Think about: *class weights*, *threshold tuning*, *stratified sampling*.

**Scenario:** An insurance company wants to predict how much a person's *medical insurance will cost* based on their profile.

- Each row is one policyholder
- Features describe age, health, and location
- Target: `charges` — annual cost in **USD**

**Why does this matter?**

Accurate pricing keeps premiums fair for healthy customers while ensuring the company stays solvent. Under-prediction loses money; over-prediction loses customers.

# Insurance Dataset — Features (6 columns)

| Feature | Type | Description |
|---|---|---|
| `age` | integer | Age of the policyholder |
| `sex` | categorical | male / female |
| `bmi` | float | Body mass index |
| `children` | integer | Number of dependents |
| `smoker` | categorical | yes / no |
| `region` | categorical | US geographic region |

### Hint

There are *non-linear interactions* hiding in these features. A simple linear model misses them — feature engineering pays off.

# Insurance Dataset — Key Statistics

| **Property** | **Value** |
|---|---|
| Training samples | 1,070 |
| Test samples | 268 |
| Number of features | 6 |
| Mean charges | $13,346 |
| Median charges | $9,575 |
| Min charges | $1,122 |
| Max charges | $62,593 |

**Watch out — Skewed Target!**

The distribution of `charges` is *heavily right-skewed*. A few high-cost patients pull the mean far above the median.

Think about: *log-transforming* the target, *polynomial features*.

# Scoring Formula

Your submission is scored by:

$$\boxed{\text{Score} = \frac{\text{Macro F1} + \max(0,\ R^2)}{2} \times 100}$$

**Macro F1** (Classification)

- Average of F1 for class 0 and F1 for class 1
- Penalizes models that ignore the minority class
- Range: 0 to 1

$R^2$ (Regression)

- How close are your predictions to the real values?
- 1.0 = perfect predictions
- 0.0 = no better than always guessing the average
- Negative $R^2$ is clipped to 0

Baseline (Logistic + Linear Regression): *~72* | Your target: *90+*

**Tip:** Use *cross-validation* (CV) to estimate your real score before submitting — don't rely only on the public leaderboard!

# Your Files & Submission

**You receive 4 CSV files:**

| File | Shape | Description |
|---|---|---|
| `train_clf.csv` | $5{,}625 \times 20$ | Classification train (has `Churn`) |
| `test_clf.csv` | $1{,}407 \times 19$ | Classification test (*no target*) |
| `train_reg.csv` | $1{,}070 \times 7$ | Regression train (has `charges`) |
| `test_reg.csv` | $268 \times 6$ | Regression test (*no target*) |

**You submit:**

- `y_pred_clf` — array of **0** or **1**, length 1,407
- `y_pred_reg` — array of **floats**, length 268
- Call `generate_submission(y_pred_clf, y_pred_reg)`
- Upload the resulting `submission.csv` to Kaggle

## Credits

# KAUST Academy

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